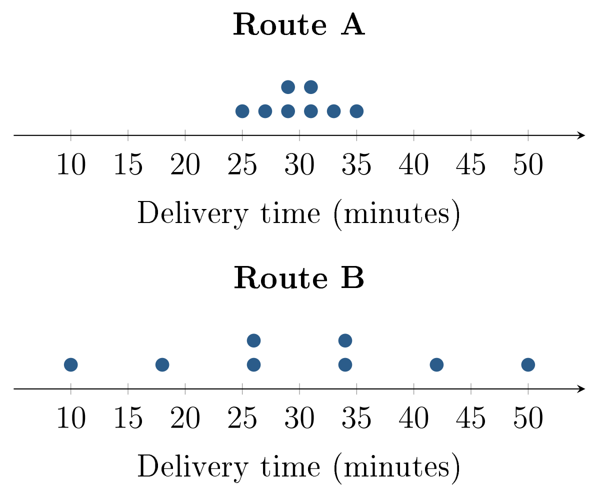
Same Mean, Different Deadline Risk

Unit 2 · Topic 2.9 · TODAY'S PROBLEM

Suppose a dispatcher models delivery time using eight equally likely outcomes for each route, shown in the dotplots. An analyst reports that both routes have mean 30 minutes, with $\sigma_A = 3$ minutes and $\sigma_B = 12$ minutes.

The dispatcher says, *The routes have the same mean, so it does not matter which one a customer gets.* A customer has 40 minutes before an appointment and places a high cost on being late.

For the eight outcomes, the sums of squared distances from 30 are 72 for Route A and 1,152 for Route B.



FIRST TAKE (5 min, on your sheet)

Which route would you assign to this customer? Cite one number or dotplot feature.

DOK 1 **(a)** Count the outcomes longer than 40 minutes for each route.

DOK 2 **(b)** Use the given sums of squared distances to verify both population standard deviations. Interpret what one of them means in minutes.

DOK 3 **(c)** Which route fits this customer's priority? Use both standard deviations and the chance of exceeding 40 minutes. Explain why the dispatcher's mean-only comparison can mislead this customer. Write 3–4 sentences.

Optional review: follow-along (scan the code)

Work (a)–(c) from the rules on your sheet. Turn in your sheet.